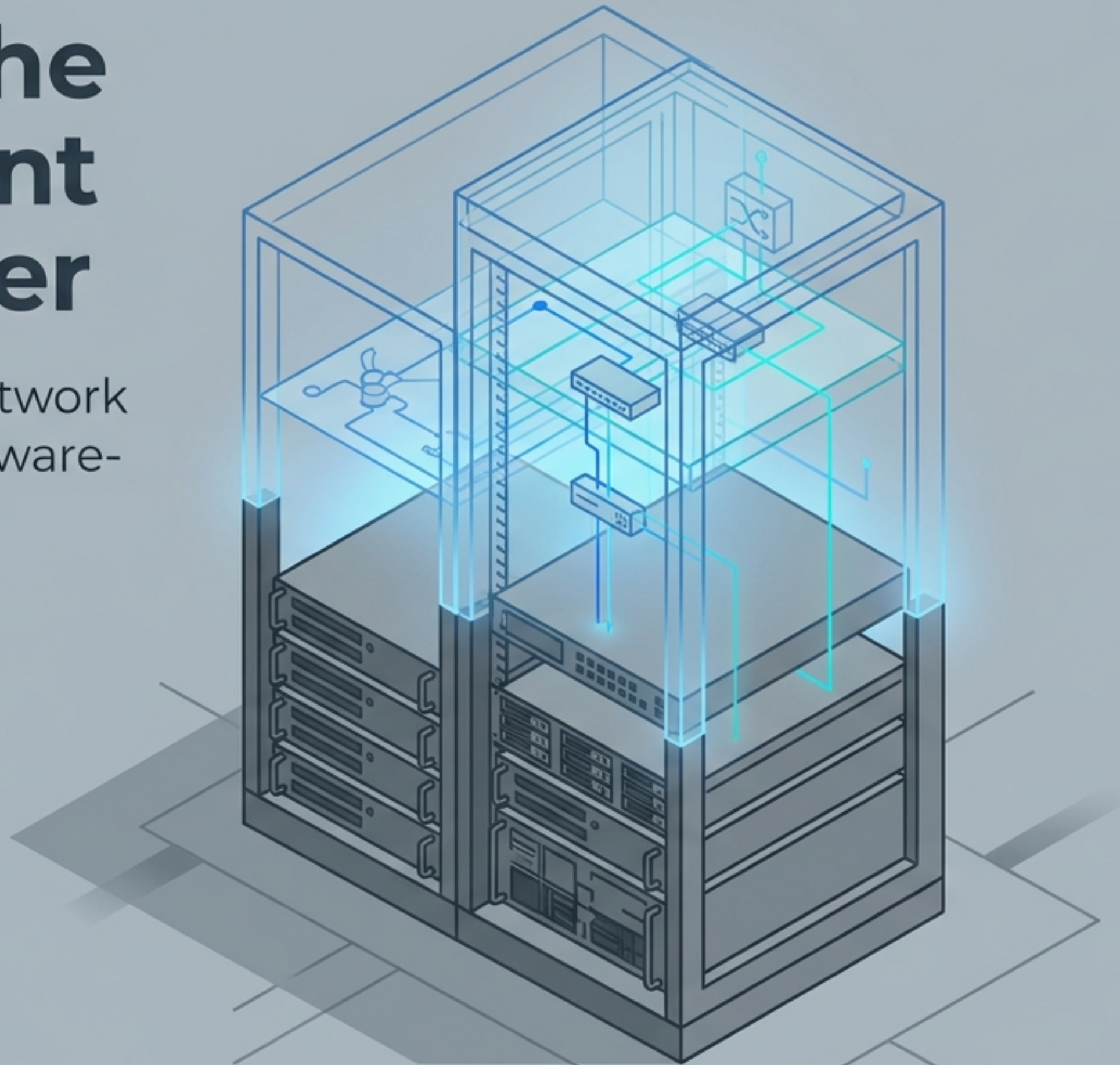


Building the Transparent Data Center

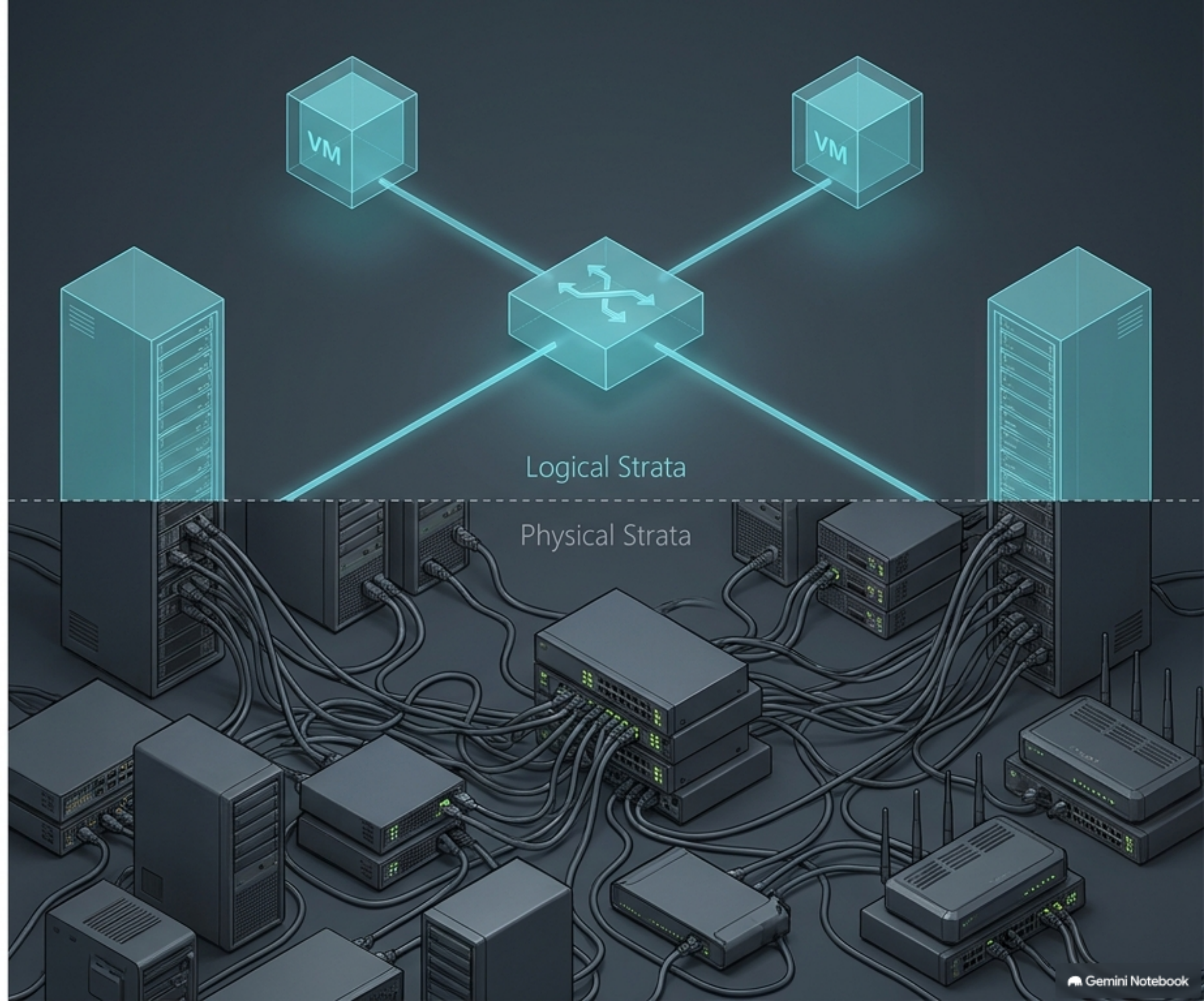
A Mental Model for Network Virtualization and Software-Defined Abstractions.



Separating logical behavior from tangible infrastructure

Network virtualization applies the same abstraction to the network that operating systems apply to RAM.

Workloads interact with logical network interfaces and topologies, while the shared physical infrastructure transparently carries the underlying traffic.



The challenge of shared physical infrastructure

A single physical network interface must serve many isolated workloads. Virtualization solves this by providing:



Isolation

Tenants share infrastructure securely without accessing each other's traffic.



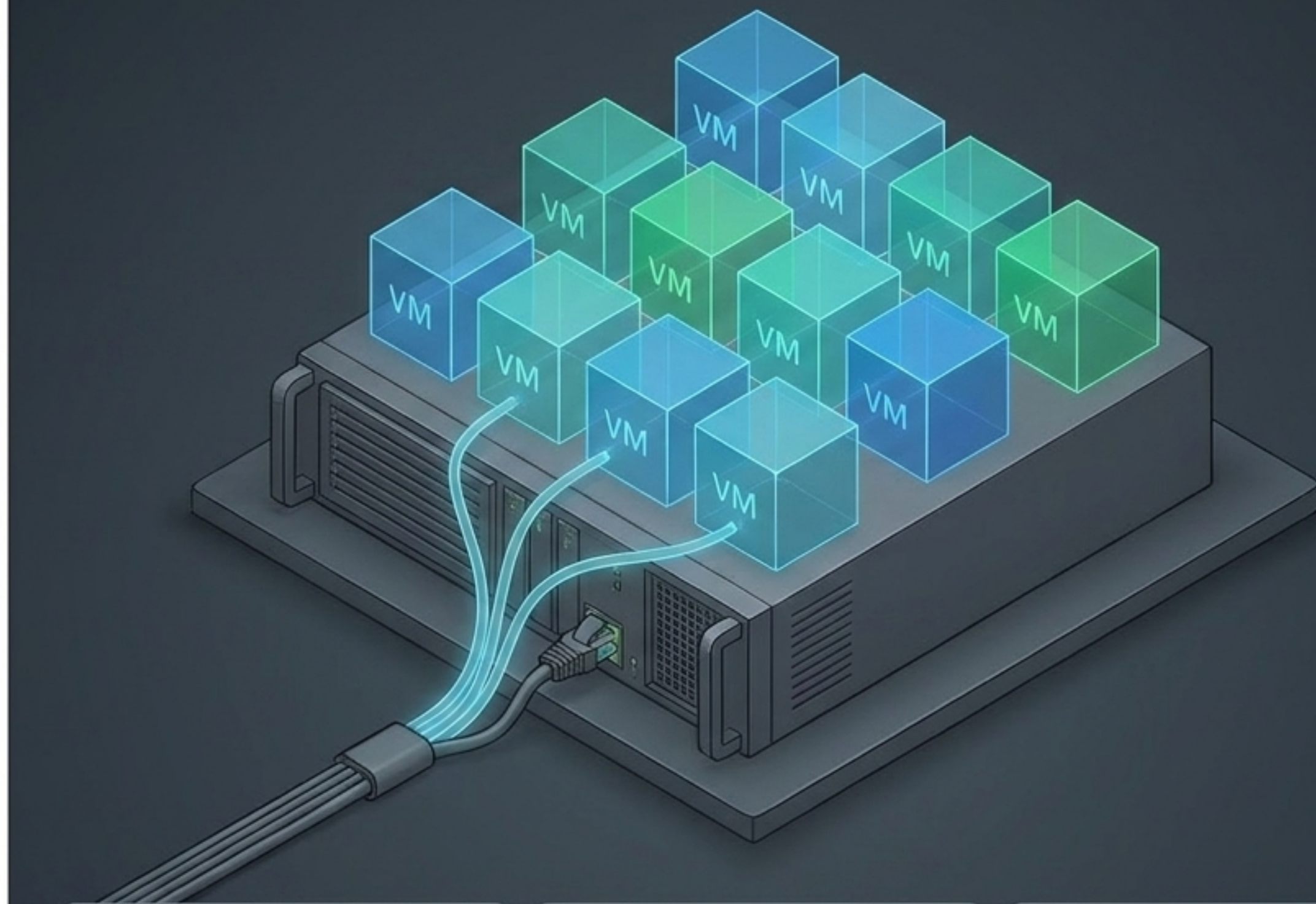
Flexibility

Logical networks are created, modified, or deleted in software without rewiring.

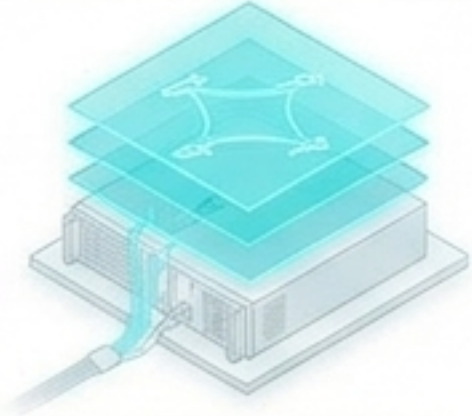


Workload Mobility

A VM can move between physical hosts while retaining its logical network identity.



Two distinct but complementary architectures



Network Virtualization (The "What")

Core Question

How do we abstract it?

Primary Focus

Creating logical networks and resources independent of physical topology.

Typical Goal

Isolate tenants and workloads over shared infrastructure.



Software-Defined Networking (The "How")

Core Question

How do we control it?

Primary Focus

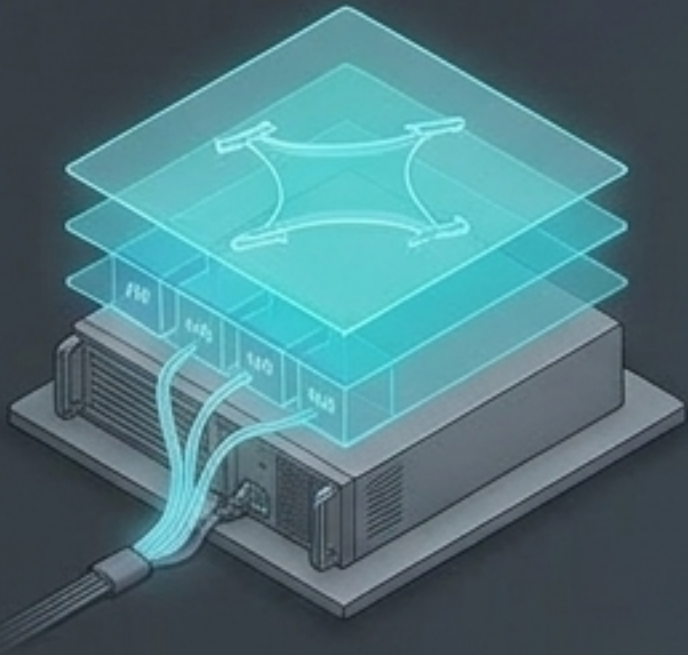
Separating network control logic from packet forwarding.

Typical Goal

Make network behavior centralized, programmable, and automated.

Four approaches to abstracting the network

Hypervisor-Based



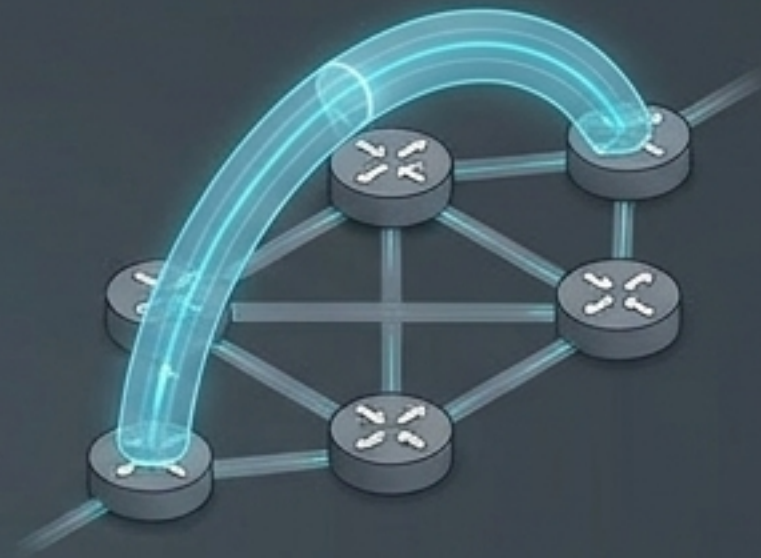
Environment:
Virtual Machines.

Software Switch-Based



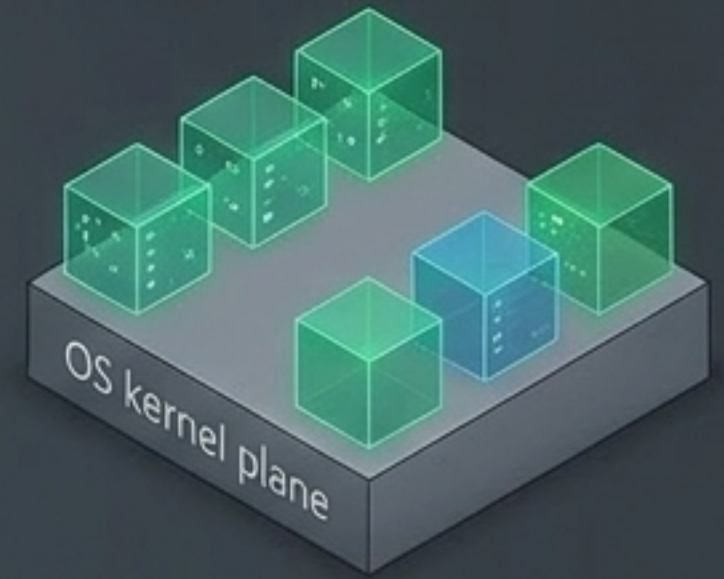
Environment:
Hosts, VMs, Containers
(e.g., Open vSwitch,
Linux Bridges).

Overlay-Based



Environment:
Cloud and Data Center
Fabrics.

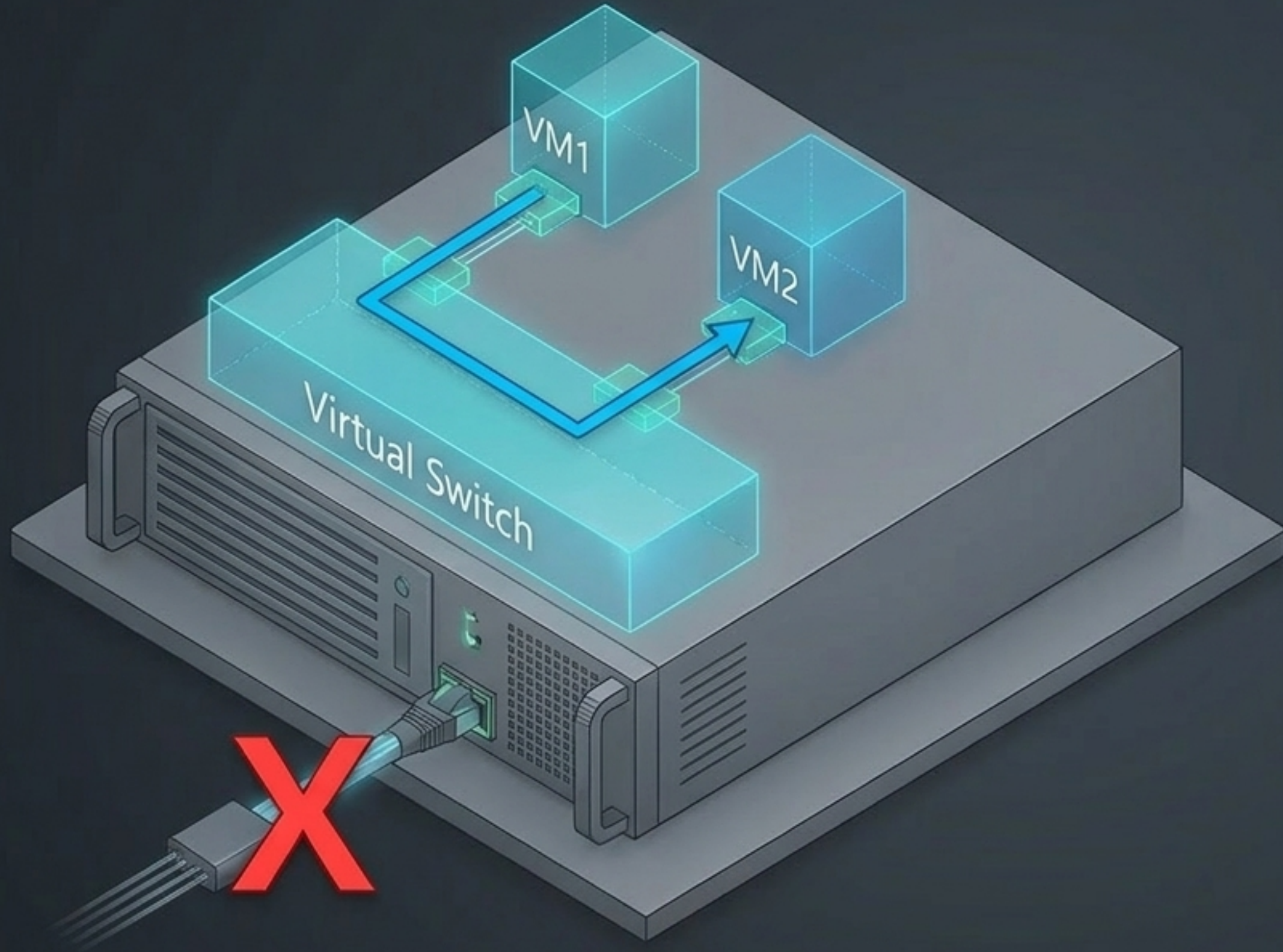
Container-Based



Environment:
Kubernetes / Docker.

These approaches are not mutually exclusive. A modern cluster can utilize namespaces, software switches, and overlays simultaneously.

Software switching keeps local traffic off the physical wire



A virtual machine behaves as though it has a physical network adapter.

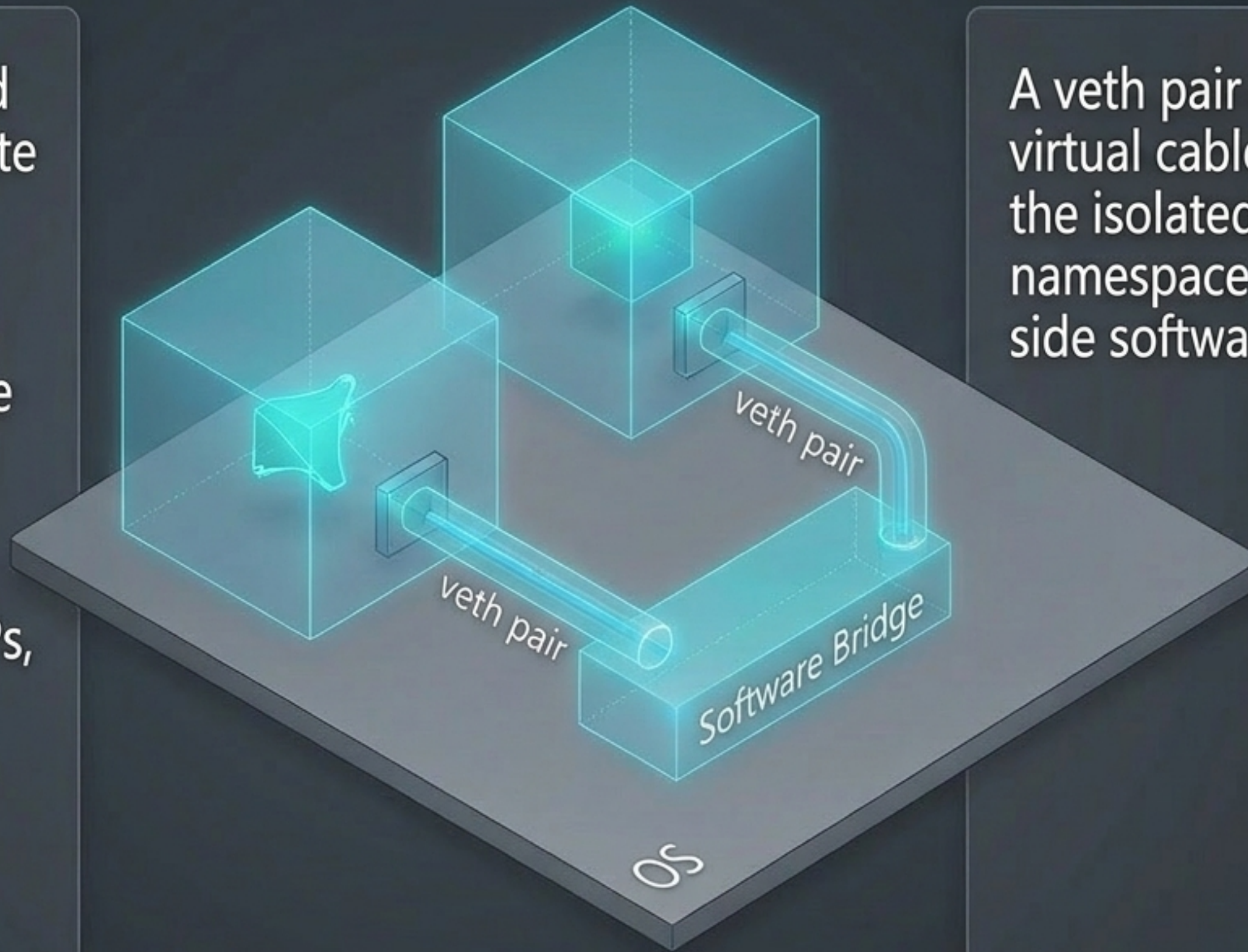
However, if VM1 communicates with VM2 on the same host, the traffic routes entirely through the hypervisor's virtual switch.

A physical switch connected to the host will never see this traffic.

Containers require isolated network namespaces

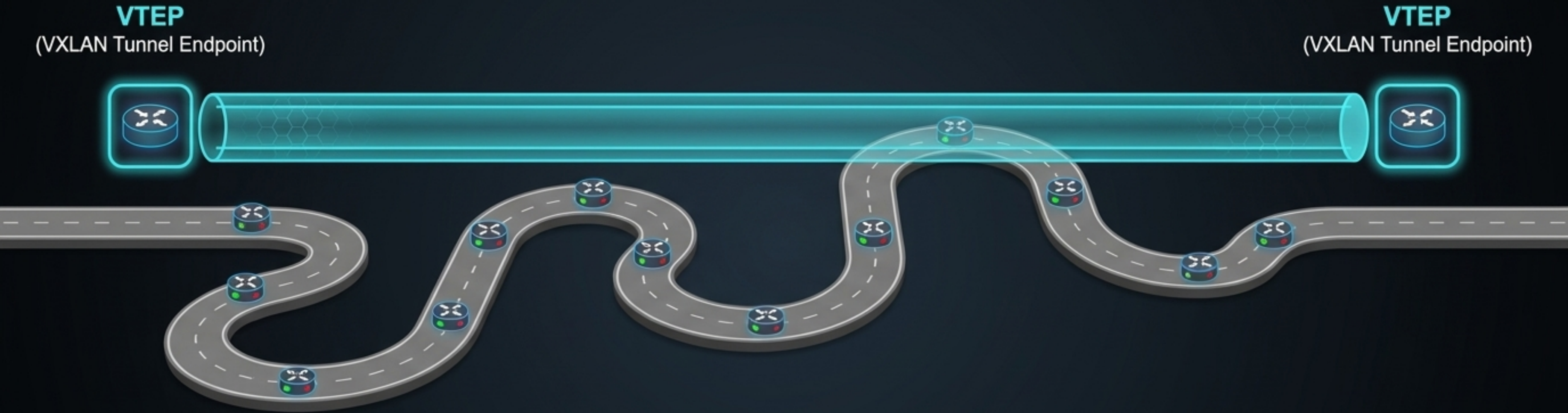
Containers are isolated processes, not complete virtual machines.

Linux network namespaces give these processes independent network environments (separate interfaces, IPs, and routing tables).



A veth pair acts as a virtual cable, connecting the isolated container namespace to a host-side software bridge.

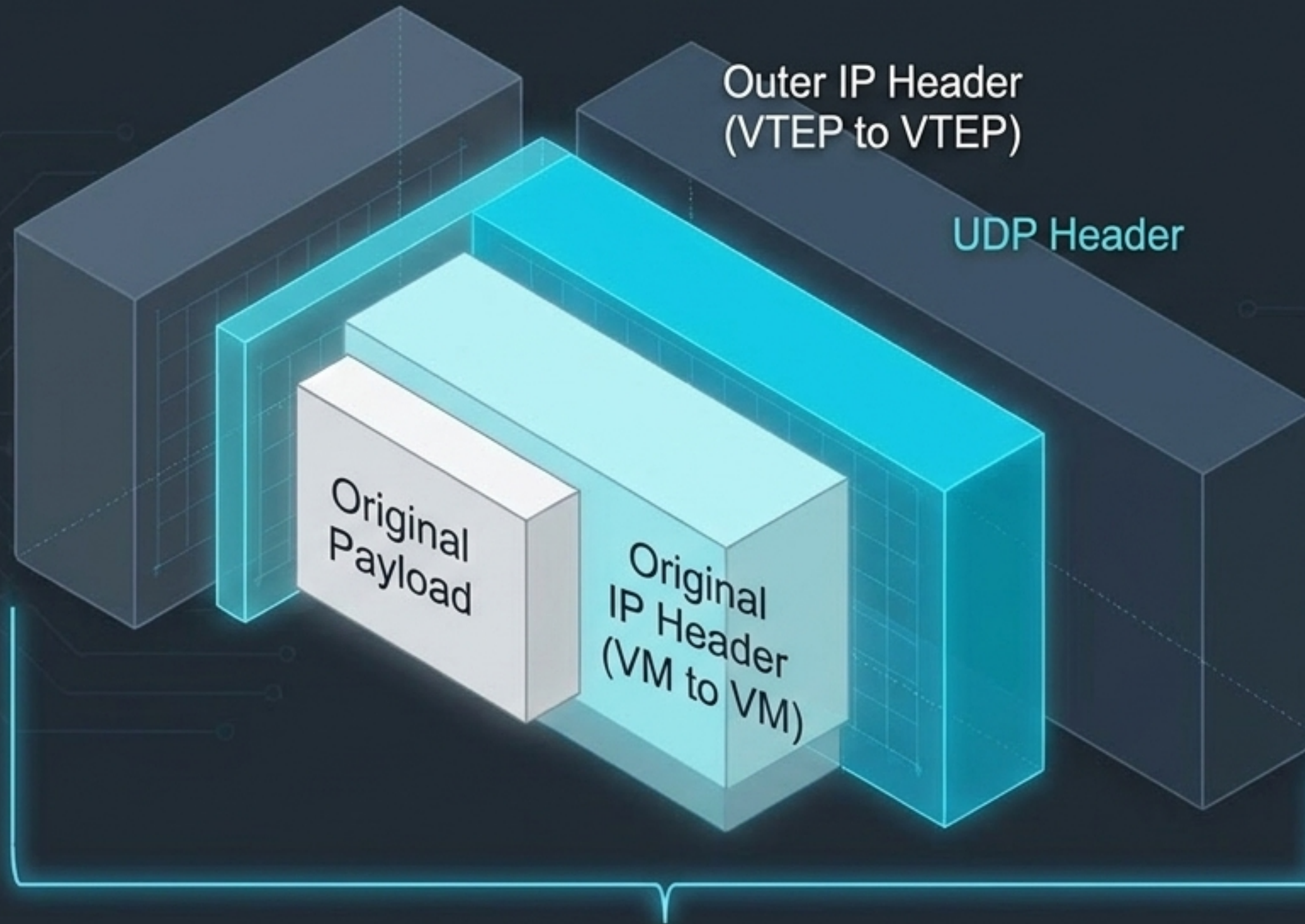
Building logical bridges over physical roads



The underlying physical network (Underlay) only needs to provide basic IP connectivity.

The logical network (Overlay) encapsulates traffic, hiding the logical tenant topology from the physical hardware. VXLAN enables massive logical segmentation across a routed IP fabric.

Encapsulation adds headers and demands larger physical frames

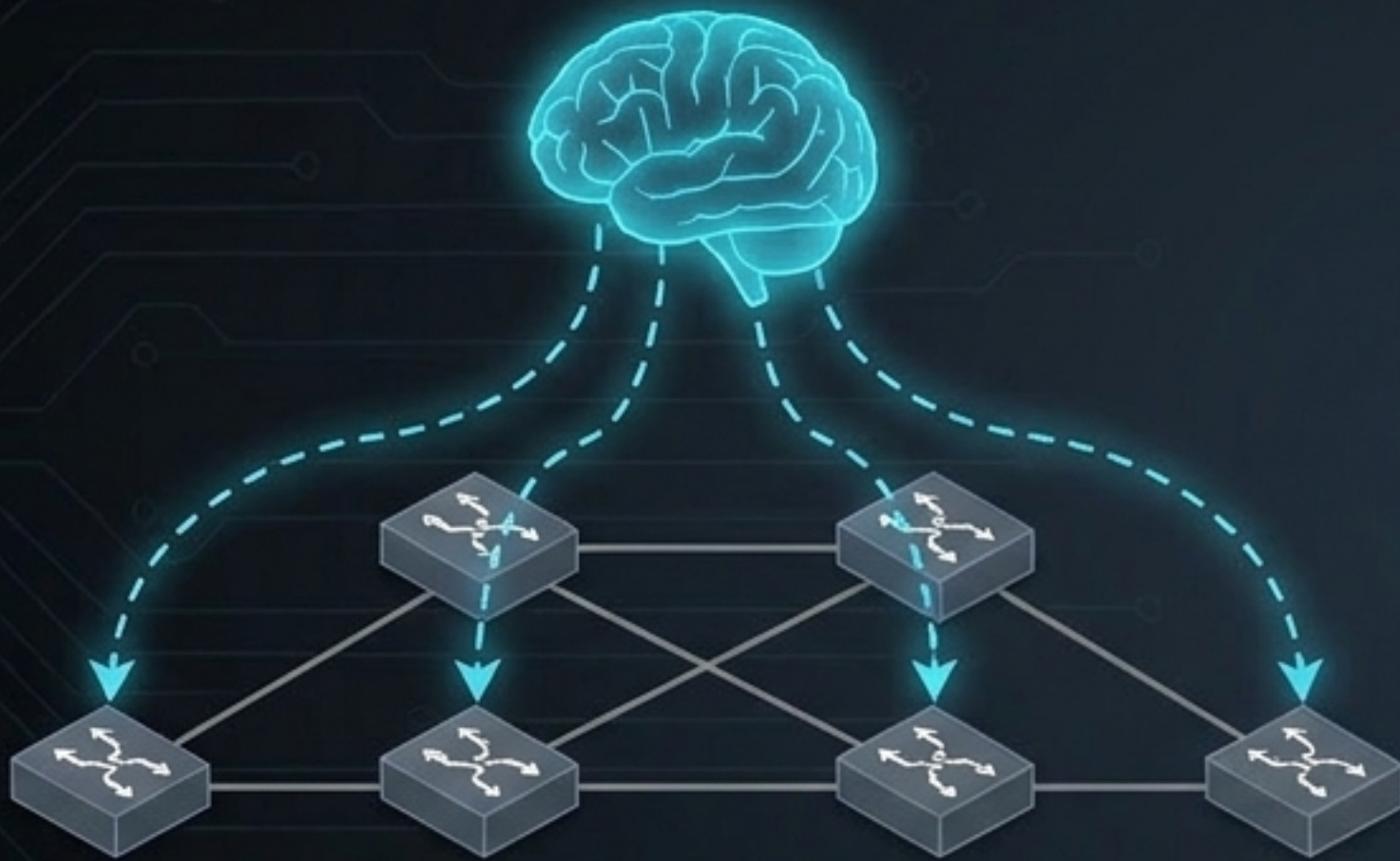


Increased MTU requirement

A packet traveling through an overlay network is encapsulated inside another protocol. This process adds headers, creating a packet larger than the original payload. If the physical network's Maximum Transmission Unit (MTU) is not adjusted to carry these larger frames, unpredictable fragmentation and packet loss will occur.

Balancing central visibility with distributed processing

Centralized Architecture



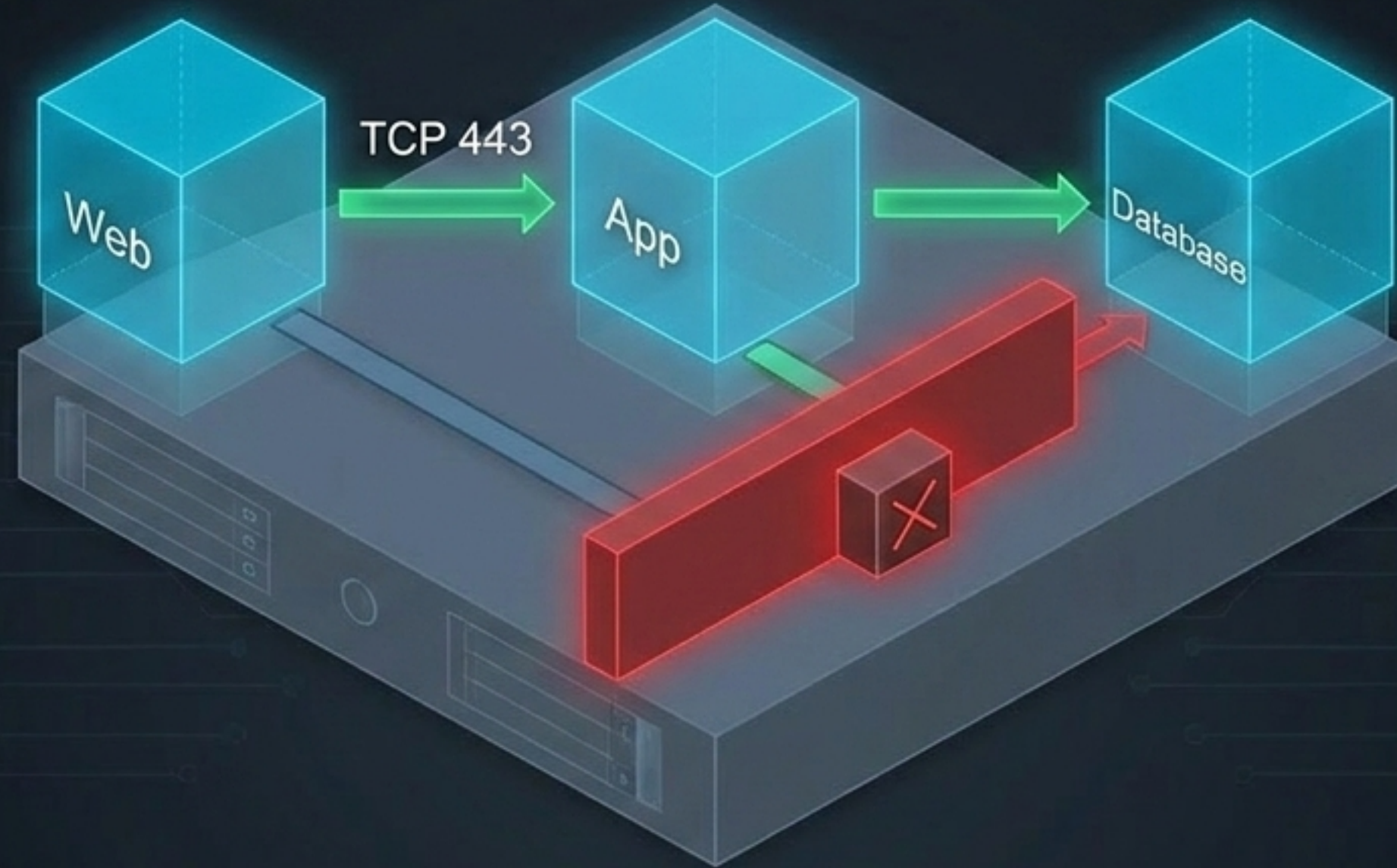
Provides unified visibility, easier automation, and consistent policy enforcement. Controllers maintain a global view of the logical network.

Distributed Architecture



Pushes packet forwarding and routing down to the individual host nodes. Eliminates centralized bottlenecks, dramatically improving scalability and performance for East-West traffic.

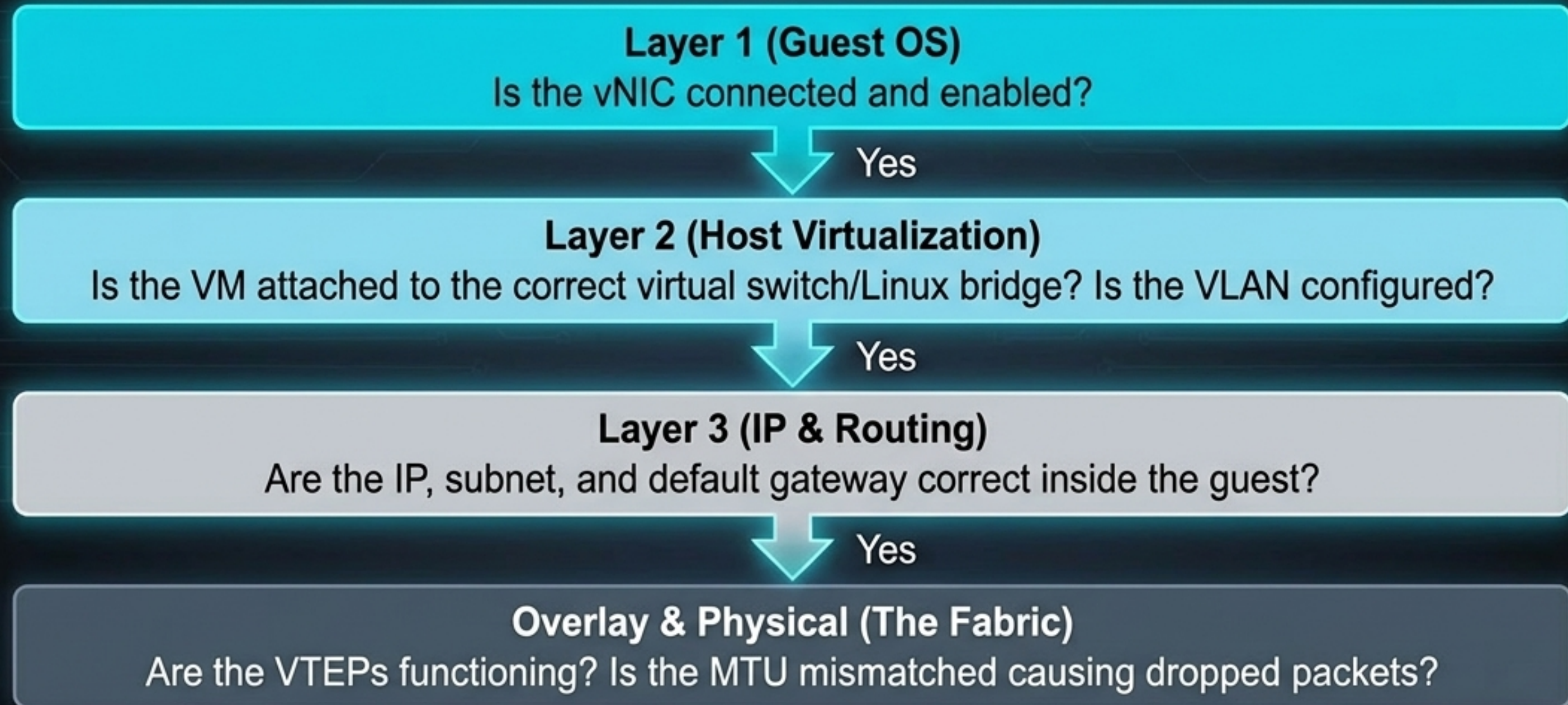
Microsegmentation enforces security at the workload boundary



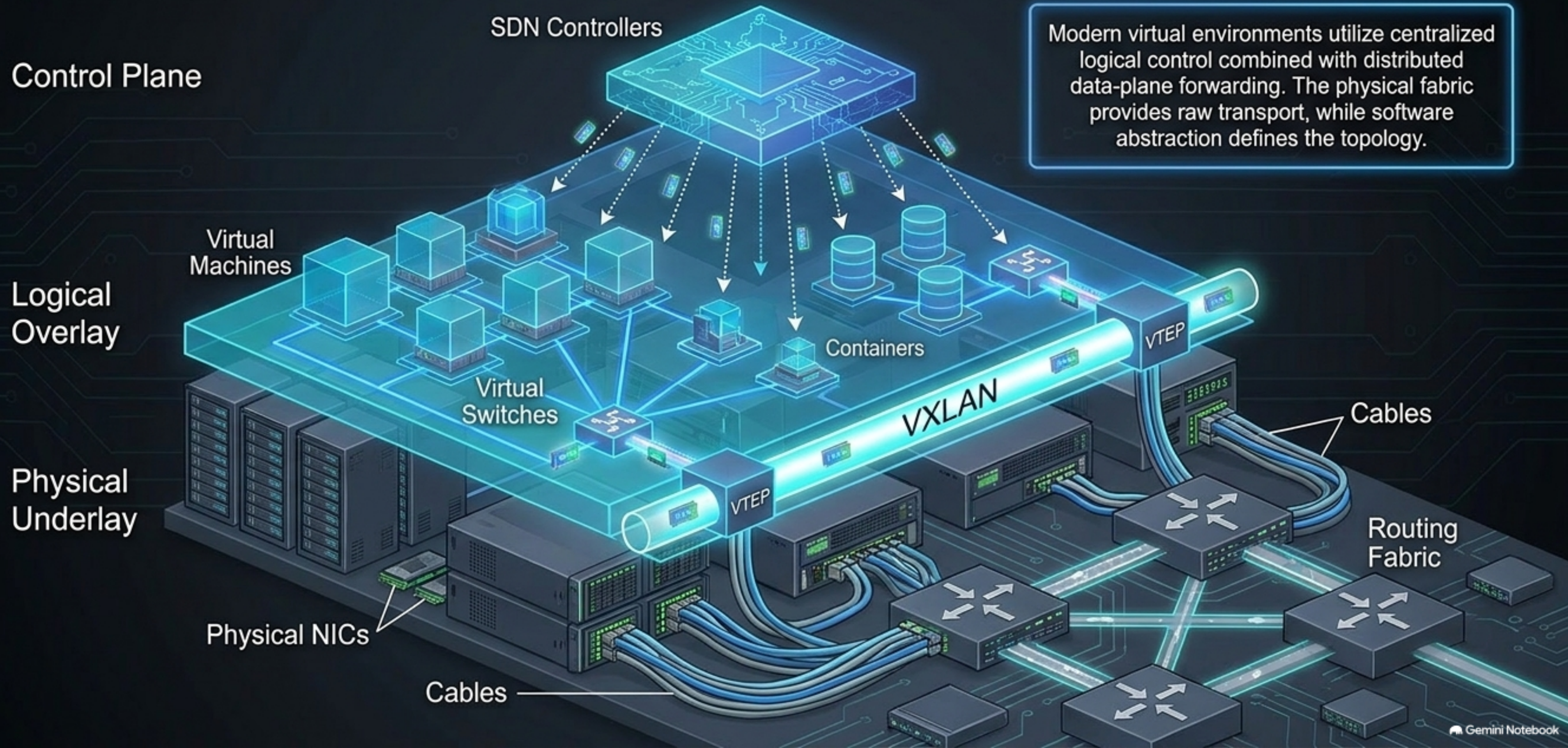
Network virtualization detaches security from rigid physical choke points. Microsegmentation applies firewall policies directly at the virtual network interface level. Even if workloads share the exact same physical host and logical subnet, unauthorized lateral movement is blocked in software before it ever reaches the physical wire.

Systematically diagnosing the virtual network

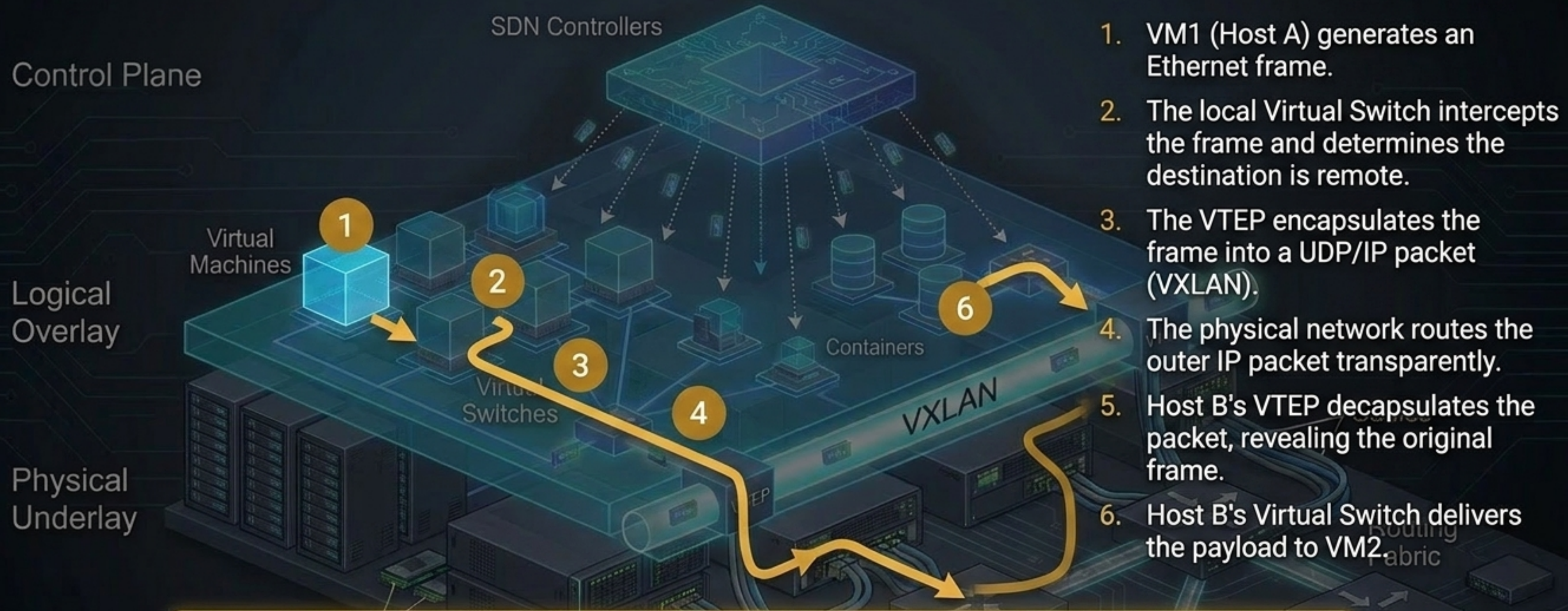
A running virtual machine is not necessarily a connected one. Troubleshooting must start at the logical source and step downward. Never immediately assume the physical switch is broken.



The Complete Architecture: Integrating the Strata



The Journey of a Packet



Key Insight: The physical network transports the overlay traffic without ever needing to understand or parse the complete logical tenant topology hidden inside it.